Chemical Bonding

What is chemical bond?

The attractive forces that hold two or more atoms together in a molecule is called bond.

Types of Chemical Bonds:

Si.	Types of Chemical Bonds	Examples
1.	Ionic bond	NaCl, KCl, $MgCl_2$
2.	Covalent bond	CH_4, NH_3, PCl_5
3.	Co-ordinate bond	NH_4^+, $[Co(NH_3)_6]^{2+}$
4.	Metallic bond	Au, Pt, Na, Cu, Fe
5.	Hydrogen bond	H_2O, HF, nitrophenol

1

Ionic Bonding

The chemical bond formed between two atoms by the complete transfer of one or more valence electrons from one atom to the other is called ionic bond. Ionic bonding results from electrostatic interactions among ions. This bond is also called **electrovalent** or **polar bond**.

$$Na\ (2,8,1) \xrightarrow[-e^-]{\text{loss of electron}} Na^+\ (2,8)$$

$$Cl\ (2,8,7) \xrightarrow[+e^-]{\text{gain of electron}} Cl^-\ (2,8,8)$$

$$Na^+\ (2,8) + Cl^-\ (2,8,8) \xrightarrow{\text{combination of ions}} Na^+Cl^-\ \text{Crystal}$$

2

Ionic Bonding

$Na(g) \longrightarrow Na^+(g) + e^-$ first ionization energy = 496 kJ/mol

$Cl(g) + e^- \longrightarrow Cl^-(g)$ electron affinity = −349 kJ/mol

$IP + EA = (496 - 349)\ kJ/mol = 147\ kJ/mol$

$Na^+(g) + Cl^-(g) \longrightarrow NaCl(s)$ crystal lattice energy = −789 kJ/mol

Crystal lattice energy = [496 + (− 349) + (-789)] kJ/mol = - 642 kJ/mol

The crystal (solid) formation thus further lowers the energy to (147 -789) kJ/mol = -642 kJ/mol. The overall result is that one mole of NaCl(s) is 642 kJ/mol lower in energy than the original mixture of atoms.

3

Ionic Bonding

The ***anions formed always achieve an inert gas configuration*** when forming ionic bond, while the ***cations formed may achieve any of the following configurations***:

- Inert gas configuration (ns^2 or ns^2p^6) Na^+, K^+, Mg^{2+}, Ca^{2+}
- Pseudo-inert gas configuration ($ns^2p^6d^{10}$) Cu^+, Ag^+, Au^+, Zn^{2+}, Cd^{2+}
- The inert pair configuration i.e., (18+2) electrons configuration [(n-1) $s^2p^6d^{10}.ns^2$] IIIA: Ga, In, Tl (ns^2p^1)
- IVA: Ge (ns^2p^2); Sn (ns^2p^2); Pb (ns^2p^2); VA: Sb (ns^2p^3); Bi (ns^2p^3):
  (+2), +4 (+2), +4 (+2), +4 (+3), +5 (+3), +5

4

2023/02/20 11:01

Ionic Bonding

Conditions for the formation of ionic compounds

1) The ionization energy of the metal atom should be low.

A (g) + Energy required (EA) ⟶ A^+ (g) + e^-
metal atom → metal ion/cation

2) Electron affinity of the non-metal should be high.

B (g) + e^- ⟶ B^- (g) + Energy released (EA)
non-metal atom → anion

3) The lattice energy of the ionic compound formed should be high.

A^+ (g) + B^- (g) ⟶ A^+B^- (crystal) + Energy released
1 mol 1 mol → 1 mol → (Lattice energy)

Ionic Bonding

***Lattice energy**: Lattice energy is the energy released when one mole of a crystalline compound is assembled at 0K from its infinitely separated components.*

Higher the value of lattice energy of a crystal, the greater is the ease of its formation, i.e., greater will be the stability of the ionic crystal.

4) The electronegativity difference between metal and non-metal should be high. In fact, *electronegativity difference of 2 or more is favorable to form an ionic bond.* The electronegativity difference between Na and Cl is 2.1 (Na = 0.9, Cl= 3.0). So Na and Cl form ionic bond in NaCl molecule.

Properties of ionic compounds

1) Most of the ionic compounds are crystalline solids.

2) They are hard , have high melting and boiling points. For example, the melting point of NaCl is 801° C and the boiling point of liquid NaCl is 1413°C.

3) They are soluble in polar solvents, e.g., H_2O, C_2H_5OH etc., but in soluble in non-polar solvent, such as CCl_4, CH_2Cl_2, benzene etc.

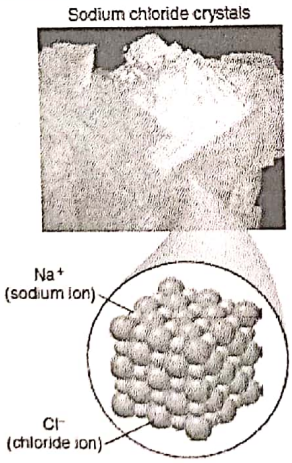


Figure: Sodium Chloride crystal

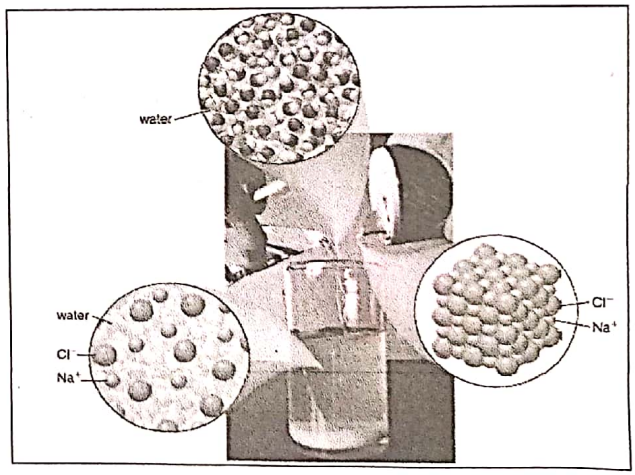
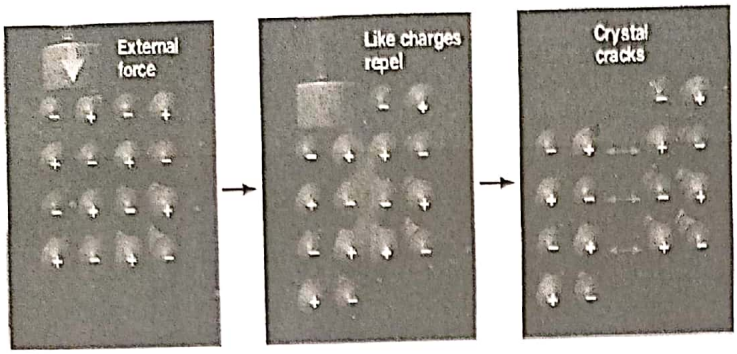


Figure: Dissolution of NaCl in water

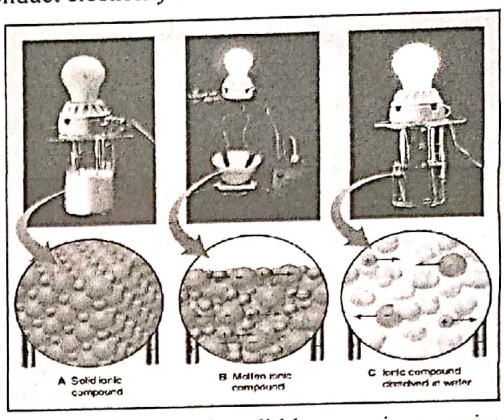
2023/02/20 11:01

4) They are highly brittle.



The positive and negative ions in the crystal are arranged to maximize their attractions. When an external force is applied, like charges move near each other and the repulsions crack the piece apart. 9

5) They conduct electricity in solution or in molten state.



A: No current flows in the ionic solid because ions are immobile. **B**: In the molten compound, mobile ions flow toward the oppositely charged electrodes and carry a current. **C**: In an aqueous solution of the compound, mobile solvated ions carry a current. 10

2023/02/20 11:01

Chemical bond

Covalent Bond

Covalent bond: A covalent bond is formed when two atoms share one or more pairs of electrons. Covalent bonding occurs when the *electronegativity difference* between atoms of the elements *is zero or relatively small.*

In 1916 Gilbert Newton Lewis proposed that the strong attractive force between two atoms in a molecule results from a covalent bond. He depicted the formation of a chemical bond in H_2.

Covalent Bond

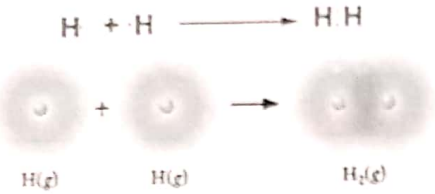


Figure: The electron probability distribution for the H_2 molecule. The electron density (shown in red) occupies the space around both atoms.

Lewis uses electron-dot symbol for the hydrogen atoms and represents the covalent bond by a pair of dots. Recall that the ***two electrons from the covalent bond spend part of the time in the region of each atom. In this sense, each atom in H_2 has a helium configuration.***

Covalent Bond

The formation of a bond between H and Cl to give an HCl molecule can be represented in a similar way.



A formula using dots to represent valence electrons is called a Lewis electron-dot formula. An electron pair represented by a pair of dots in such a formula is either a bonding pair *(an electron pair shared between two atoms)* or a lone, or nonbonding, pair *(an electron pair that remains on one atom and is not shared)*. For example,

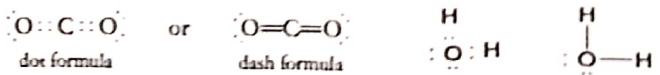


Covalent Bond

Bonding pairs are often represented by dashes rather than by pairs of dots.



The structures we use to represent covalent compounds, such as H_2 and F_2, are called Lewis structures. ***A Lewis structure is a representation of covalent bonding in which shared electron pairs are shown either as lines or as pairs of dots between two atoms and lone pairs are shown as pairs of dots on individual atoms.*** Only valence electrons are shown in a Lewis structure.



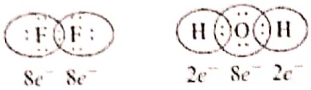
2023/02/20 11:01

The Octet Rule

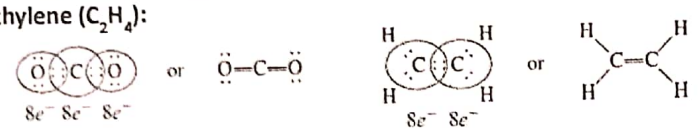
Many Lewis formulas are based on the idea that in most of their compounds, the representative elements achieve noble gas configurations (i.e., ns^2np^6 configuration)

This statement is usually called the octet rule, because the noble gas configurations have 8 e^- in their outermost shells.

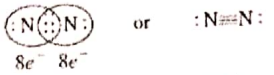
In the F_2 and H_2O molecules, the F and O atoms achieve a noble gas configuration by sharing electrons:



Double bonds are found in molecules of carbon dioxide (CO_2) and ethylene (C_2H_4):



A triple bond arises when two atoms share three pairs of electrons, as in the nitrogen molecule (N_2):



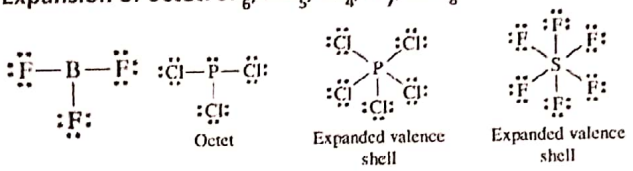
The acetylene molecule (C_2H_2) also contains a triple bond, in this case between two carbon atoms:



Limitations of the octet rule (Lewis concept)

(i) Incomplete octet: $BeCl_2$, BF_3, $(CH_3)_3Ga$, NO, $AlCl_3$

(ii) Expansion of octet: SF_6, PCl_5, SF_4, IF_7, OsF_8



Comparison of the properties of ionic and covalent compounds

	Ionic compounds	Covalent compounds
Physical state	Crystalline solid at rt; never liquid or gases under ordinary condition.	Normally exist as gases or liquid; High molecular compound can exist as solid , e.g., SiC, AlN, diamond
Crystal structure	Three dimensional arrangement	(a) Giant molecule; diamond (b) Layer structure; graphite, CdI_2
Hardness and brittleness	Hard and brittle	Solid compounds are soft and waxy
Melting and boiling points	High melting and boiling points	Low melting and boiling points Exception: giant molecules
Solubility	Polar solvent	Non-polar solvent
Electrical conductivity	In solution and molten state, they are conductive	Poor conductors

Theories of covalent bonding

Two quantum mechanical theories are used to describe covalent bond formation and the electronic structure of molecules.

1)Valence bond theory (VBT)

2)Molecular orbital theory (MOT)

3)Valence bond theory

In chemistry, VBT is one of two basic theories, along with molecular orbital theory, that were developed to use the methods of quantum mechanics to explain chemical bonding. This theory primarily focuses on the ***formation of individual bonds*** from the atomic orbitals of the participating atoms ***during the formation of a molecule.***

2023/02/20 11:01

Theories of covalent bonding

2) Molecular orbital theory

In chemistry, molecular orbital theory is a method for describing the electronic structure of molecules using quantum mechanics. ***Electrons are not assigned to individual bonds between atoms***, but are treated as ***moving under the influence of the nuclei*** in the whole molecule.

Theories of covalent bonding

Postulates of VBT

- The overlapping of **two half-filled valence orbitals of two different atoms** results in the formation of the covalent bond. The overlapping causes the electron density between two bonded atoms to increase. This gives the property of stability to the molecule.
- In case the atomic orbitals possess ***more than one unpaired electron, more than one bond can be formed*** and ***electrons paired*** in the valence shell ***cannot take part in*** such a ***bond formation***.

Theories of covalent bonding

Postulates of VBT

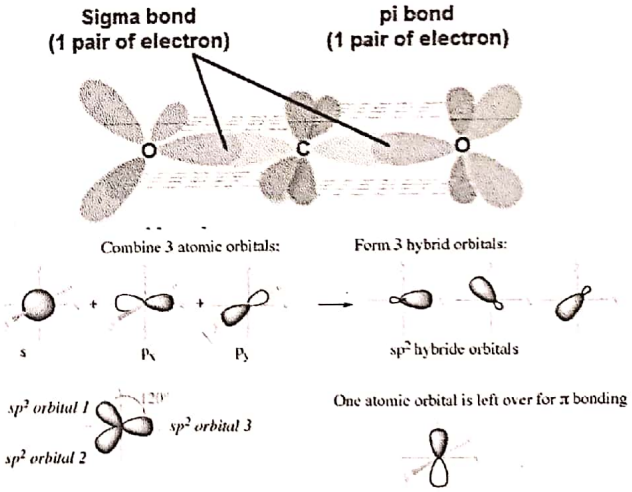
- A covalent bond is ***directional***. Such a bond is also ***parallel to*** the region of ***overlapping atomic orbitals.***
- Based on the pattern of overlapping, there are two types of covalent bonds:

(i)sigma bond
ii)pi bond.

The covalent bond formed by sidewise overlapping of atomic orbitals is known as pi (π) bond.

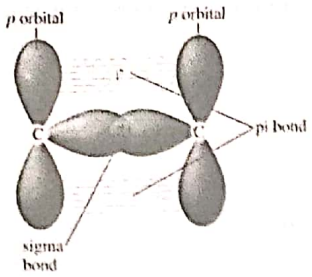
The bond formed by overlapping of atomic orbital along the inter nucleus axis is known as a sigma (σ) bond.

The formation of sigma and pi bonds is illustrated below.



2023/02/20 11:02

Theories of covalent bonding



Theories of covalent bonding

Formation of covalent bond in hydrogen molecule:



Figure: As two H atoms approach each other, their 1s orbitals begin to interact and each electron begins to feel the attraction of the other proton. Gradually, the electron density builds up in the region between the two nuclei (red color). Eventually, a stable H_2 molecule is formed when the internuclear distance is 74 pm.

Theories of covalent bonding

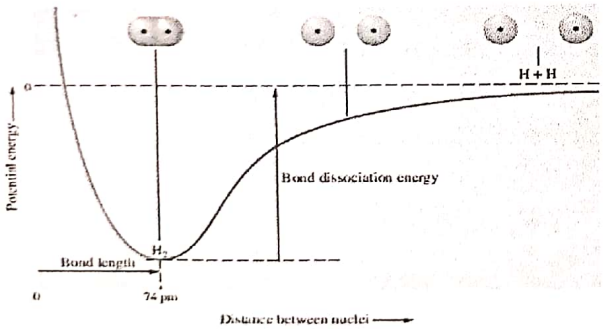


Figure: The potential energy of the H_2 molecule as a function of the distance between the two nuclei. The lowest point in the curve, -435 kJ/mol, corresponds to the internuclear distance actually observed in the H_2 molecule, 0.74 Å. (The minimum potential energy, -435 kJ/mol, corresponds to the value of -7.23×10^{-19} joule per H_2 molecule.) Energy is compared with that of two separated hydrogen atom.

Theories of covalent bonding

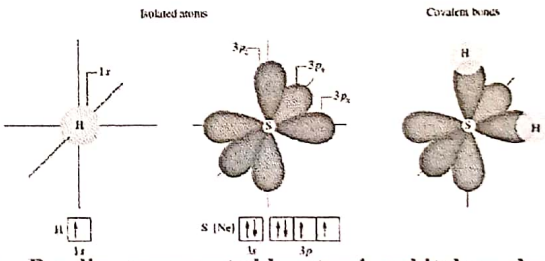


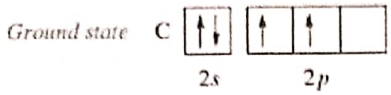
Figure: Bonding represented by atomic orbital overlapping

Figure shows the imagined overlap of atomic orbitals in the formation of **H**-to-**S** bonds in H_2S. Note that maximum overlap between the orbital of a H atom and a S atom occurs along a line joining the centers of the H and S atoms. The two half-filled sulfur 3p orbitals that overlap in H_2S are perpendicular to each other and the valence bond method suggests a bond angle of H-S-H 90°. This is in good agreement with the observed angle of 92°.

2023/02/20 11:02

Theories of covalent bonding

The ground-state electron configuration of the valence shell of carbon

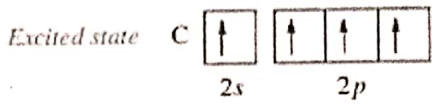


and employing only half-filled orbitals, we expect the existence of a molecule with the formula CH_2 and a bond angle of 90°. The CH_2 molecule is a highly reactive molecule observed only under specially designed circumstances.

The simplest hydrocarbon observed under normal laboratory conditions is methane, CH_4. This is a stable, unreactive molecule with a molecular formula consistent with the octet rule of the Lewis theory.

Theories of covalent bonding

The electron configuration of C atom in the *excited state* is



The electron configuration of this excited state suggests a molecule with three mutually perpendicular C-H bonds based on the 2*p* orbitals of the C atom (90° bond angles). The fourth bond would be directed to whatever position in the molecule could accommodate the fourth H atom. This description, however, does not agree with the experimentally determined H-C-H bond angles, all four of which are found to be the same as predicted by **VSEPR** theory.

Theories of covalent bonding

An algebraic combination of the wave equations of the 2s and three 2p orbitals of the carbon atom produces a new set of four identical orbitals. These new orbitals, which are directed in a tetrahedral fashion, have energies that are intermediate between those of the 2s and 2p orbitals.

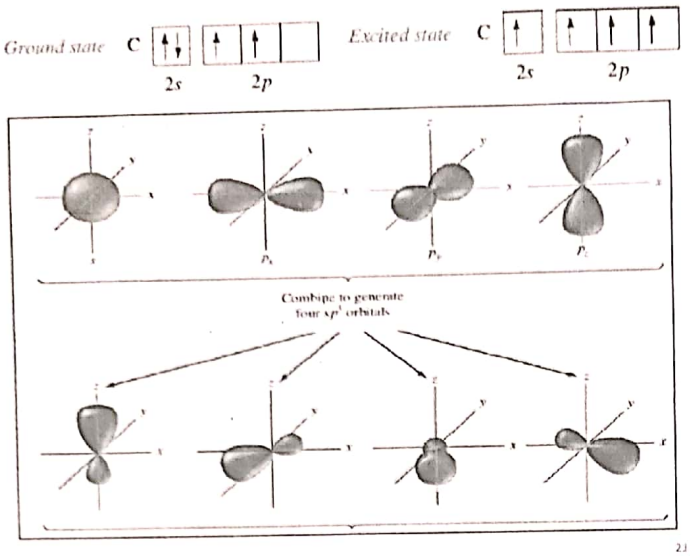
This mathematical process of replacing pure atomic orbitals with reformulated atomic orbitals for bonded atoms is called hybridization and the new orbitals are called hybrid orbitals.

Theories of covalent bonding

In a hybridization scheme, *the number of hybrid orbitals equals the total number of atomic orbitals that are combined.* Hybridization occurs only when the bonds are being formed.

The pictures in next page show the hybridization of one *s* and three *p* orbitals into a new set of four hybrid orbitals. Each sp^3 hybrid orbital has 25% *s* character and 75% *p* character.

2023/02/20 11:02



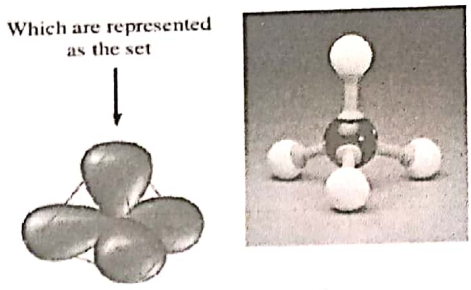


Figure : Ball-and-stick model of methane, CH_4. The molecule has a tetrahedral structure, and the H-C-H bond angles are 109.28´.

Figure: The sp^3 hybridization scheme

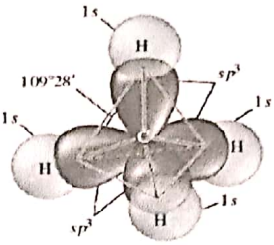
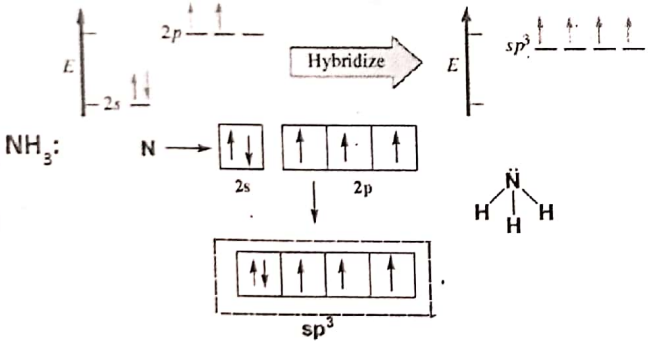


Figure: Bonding and structure of CH_4. The four carbon orbitals are hybrid orbitals (blue). Those of the hydrogen atoms (yellow) are 1s. The structure is tetrahedral, with H-C-H bond angles of 109.5°.

A useful representation of sp^3 hybridization of the valence-shell orbitals of carbon is



sp^3: CH_4, H_2O, NH_3, SO_4^{2-}, NH_4^+ ; sp^2 : BCl_3, C_2H_4, NO_3^-, $SnCl_2$; sp: $BeCl_2$, C_2H_2, CO_2; sp^3d: PCl_5, SF_4, ClF_3, XeF_2; sp^3d^2: SF_6, IF_5, XeF_4

sp^2 hybrid orbitals

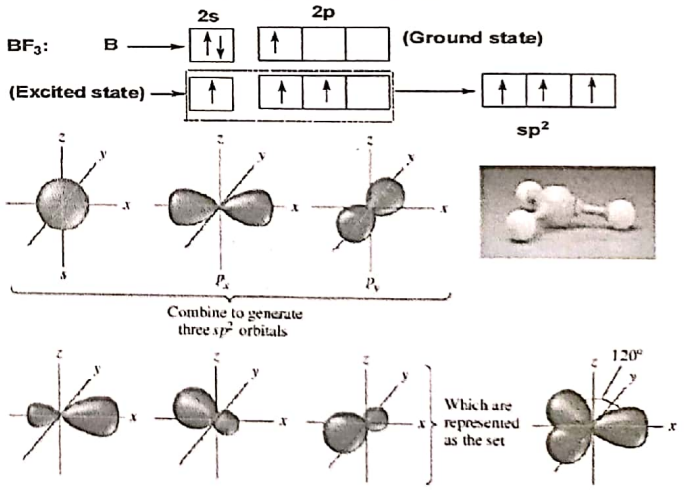


Figure : The sp^2 hybridization scheme

$H = \frac{1}{2}(V + M - C + A)$

2023/02/20 11:02

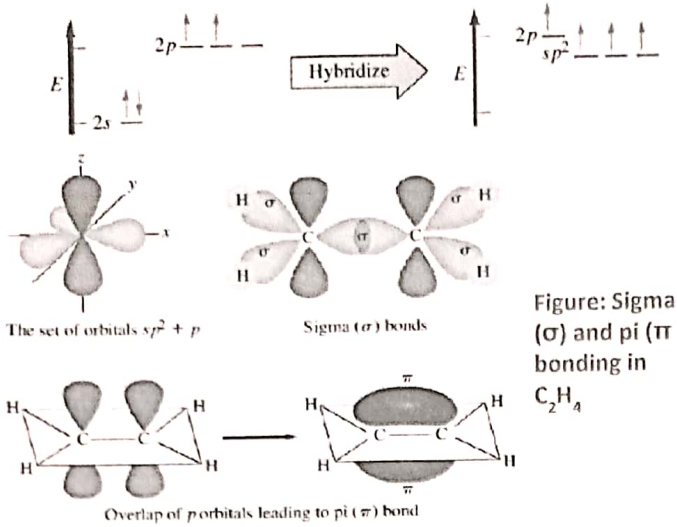


Figure: Sigma (σ) and pi (π) bonding in C_2H_4

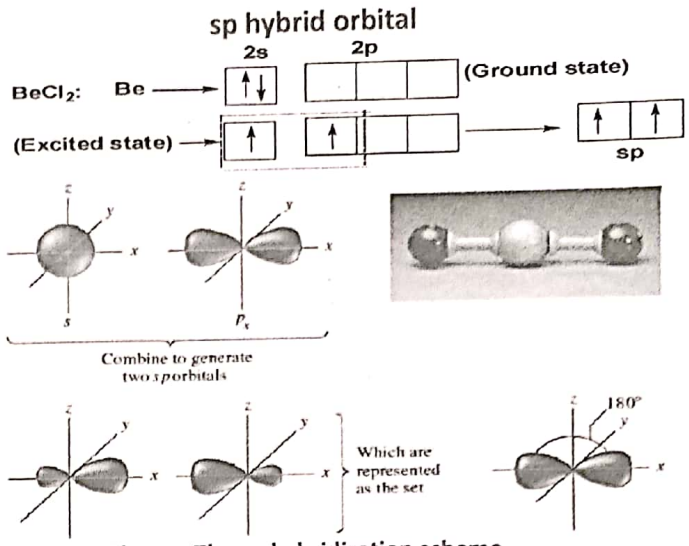


Figure : The sp hybridization scheme

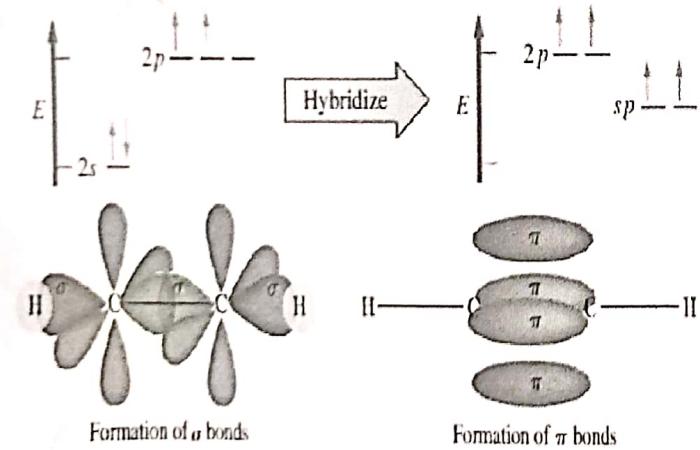
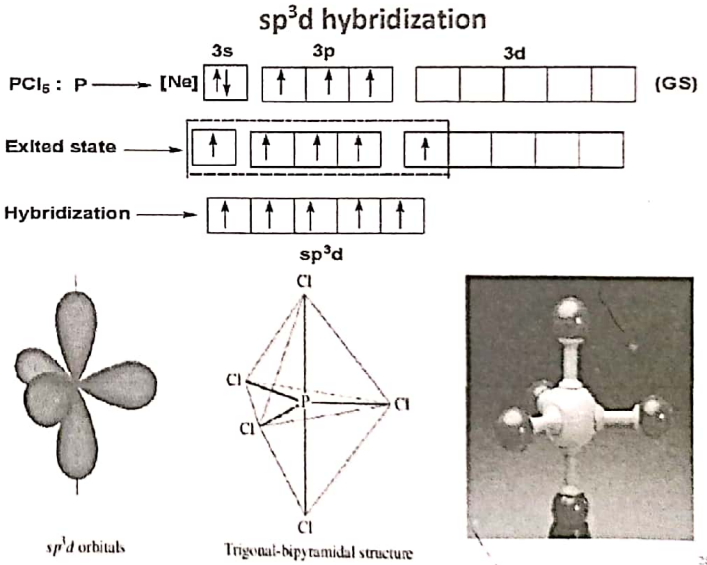
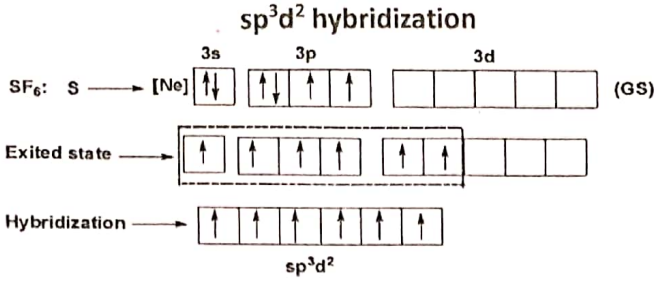


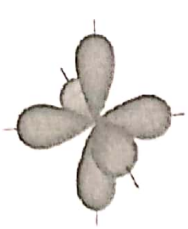
Figure: Bonding in acetylene, C_2H_2



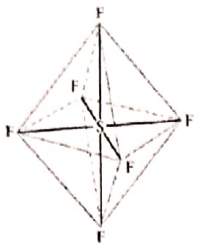
20

2023/02/20 11:02

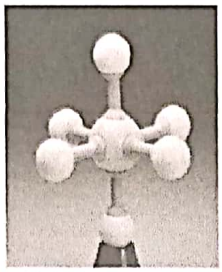




b) sp^3d^2 orbitals



Octahedral structure



Question 1: Based on the overlapping of orbitals, how many types of covalent bonds are formed and what are they?

Answer: Based on the overlapping of orbitals, two types of covalent bonds are formed. These are known as sigma(σ) and pi(π) bonds.

Sigma bonds are formed by the ***end-to-end*** overlap of atomic orbitals along the inter-nuclear axis known as a ***head-on or axial overlap***. End-on overlapping is of three types, they are s-s overlapping, s-p overlapping and p-p overlapping.

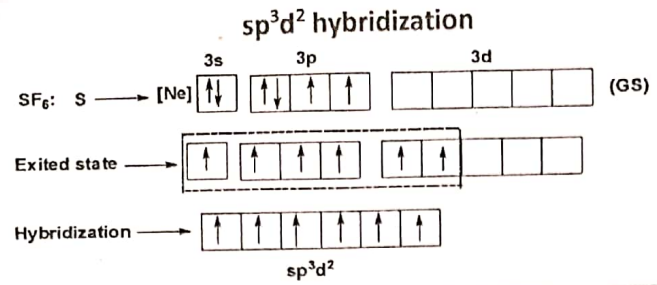
A pi bond is formed when atomic orbitals overlap in a specific way that their axes remain parallel to each other and perpendicular to the internuclear axis.

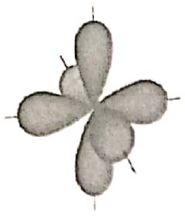
Regions of Electron Density	Arrangement		Hybridization	
2		linear	*sp*	180°
3		trigonal planar	sp^2	120°
4		tetrahedral	sp^3	109.5°
5		trigonal bipyramidal	sp^3d	90° 120°
6		octahedral	sp^3d^2	90° 90°

Molecule Type	Shape	Electron arrangement[1]	Geometry[1]	Examples
AX_2E_0	Linear			$BeCl_2$, $HgCl_2$, CO_2
AX_3E_0	Trigonal planar			BF_3, CO_3^{2-}, NO_3^-, SO_3
AX_4E_0	Tetrahedral			CH_4, PO_4^{3-}, SO_4^{2-}, ClO_4^-, $TiCl_4$, XeO_4
AX_5E_0	Trigonal bipyramidal			PCl_5
AX_6E_0	Octahedral			SF_6, WCl_6

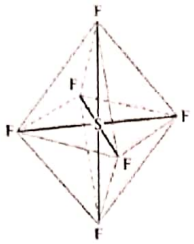
$$H = \frac{1}{2}(V + M - C + A)$$

2023/02/20 11:02

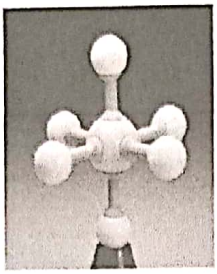




b) sp^3d^2 orbitals



Octahedral structure



29 29

Question 1: Based on the overlapping of orbitals, how many types of covalent bonds are formed and what are they?

Answer: Based on the overlapping of orbitals, two types of covalent bonds are formed. These are known as sigma(σ) and pi(π) bonds.

Sigma bonds are formed by the ***end-to-end*** overlap of atomic orbitals along the inter-nuclear axis known as a ***head-on or axial overlap***. End-on overlapping is of three types, they are s-s overlapping, s-p overlapping and p-p overlapping.

A pi bond is formed when atomic orbitals overlap in a specific way that their axes remain parallel to each other and perpendicular to the internuclear axis.

30

Regions of Electron Density	Arrangement		Hybridization	
2		linear	*sp*	180°
3		trigonal planar	sp^2	120°
4		tetrahedral	sp^3	109.5°
5		trigonal bipyramidal	sp^3d	90° 120°
6		octahedral	sp^3d^2	90° 90°

31

Molecule Type	Shape	Electron arrangement†	Geometry‡	Examples
AX_2E_0	Linear			$BeCl_2$, $HgCl_2$, CO_2
AX_3E_0	Trigonal planar			BF_3, CO_3^{2-}, NO_3^-, SO_3
AX_4E_0	Tetrahedral			CH_4, PO_4^{3-}, SO_4^{2-}, ClO_4^-, $TiCl_4$, XeO_4
AX_5E_0	Trigonal bipyramidal			PCl_5
AX_6E_0	Octahedral			SF_6, WCl_6

32

$$H = \frac{1}{2}(V + M - C + A)$$

2023/02/20 11:02

MOT

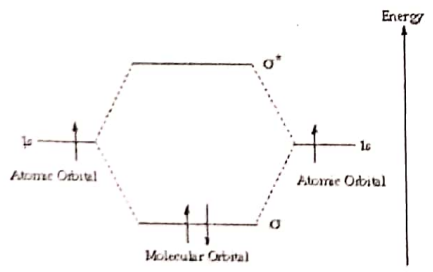
Molecular orbital theory

Lewis structures, VSEPR theory and the valence-bond theory can not provides an explanation of the electronic spectra of molecules, why oxygen is paramagnetic, or why He_2^+ is a stable species.

To address these questions, molecular orbital theory is needed to describe chemical bonding. The theory assigns the electrons in a molecule to a series of orbitals that belong to the molecule as a whole. These are called molecular orbitals.

Molecular orbital theory

According to the molecular orbital theory, individual atoms combine to form molecular orbitals. Thus the electrons of an atom are present in various atomic orbitals and are associated with several nuclei.



Molecular orbital theory

Basic ideas concerning molecular orbitals

- The number of molecular orbitals (MO) formed is equal to the number of atomic orbitals (AO) combined.
- When two atomic orbitals are combined, two molecular orbitals are formed, one is a bonding molecular orbitals (BMO) at a lower energy than the original atomic orbitals. The other is an anti-bonding molecular orbitals (ABMO) at a higher energy.
- In ground-state configurations, electrons enter the lowest energy MO available.
- The maximum number of electrons in a given MO is two.
- In ground-state configurations, electrons enter MO of identical energies singly before they pair up.

$$\textbf{Bond order} = \frac{\textbf{no. of electrons in BMO - no. of electrons in ABMO}}{2}$$

Bond order is the number of chemical **bonds** between a pair of atoms.

Molecular orbital theory

We know that we can consider electrons as either ***particle*** or ***wave nature***. Therefore, we can describe an electron in an atom as occupying an atomic orbital or by a wave function Ψ.

These are solutions to the Schrodinger wave equation. Electrons in a molecule occupy molecular orbitals. We can obtain the wave function of a molecular orbital by linear combination (addition or subtraction) of atomic orbitals.

Molecular Orbital Theory

Linear Combination of Atomic Orbitals (LCAO)

As per this method, the formation of orbitals is because of Linear Combination (addition or subtraction) of atomic orbitals which combine to form the molecule. Consider two atoms A and B which have atomic orbitals described by the wave functions Ψ_A and Ψ_B.

If the electron cloud of these two atoms overlaps, then we can obtain the wave function for the molecule by a linear combination of the atomic orbitals Ψ_A and Ψ_B. The below equation forms two molecular orbitals.

$$\Psi_{MO} = \Psi_A \pm \Psi_B$$

Molecular orbital theory

Bonding Molecular Orbitals

When the addition of wave function takes place, the type of molecular orbitals formed are Bonding Molecular Orbitals. We can represent them by

$$\Psi_{MO} = \Psi_A + \Psi_B$$

They have lower energy than atomic orbitals involved.

Anti-Bonding Molecular Orbitals

When molecular orbital forms by the subtraction of wave function, the type of molecular orbitals formed are antibonding Molecular Orbitals. We can represent them as

$$\Psi_{MO} = \Psi_A - \Psi_B$$

They have higher energy than atomic orbitals. Therefore, the combination of two atomic orbitals results in the formation of two molecular orbitals.

They are the bonding molecular orbital (BMO) and the anti-bonding molecular orbital (ABMO).

Molecular orbital theory

Relative energies of molecular orbitals

Bonding Molecular Orbitals (BMO)

Energy of bonding molecular orbitals is less than that of anti bonding molecular orbitals. This is because of the increase in the attraction of both the nuclei for both the electron of the combining atom.

Anti-Bonding Molecular Orbitals (ABMO)

Energy of anti bonding molecular orbitals is higher than bonding molecular orbitals. This is because the electron tries to move away from the nuclei and are in a repulsive state.

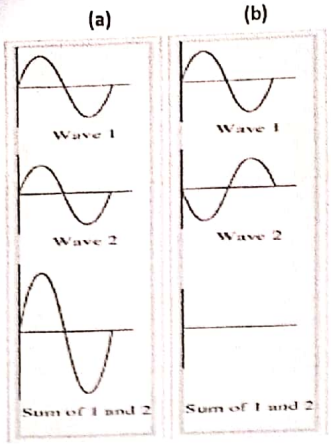


Figure: Constructive interference (a) and destructive interference (b) of two waves of the same wavelength and amplitude.

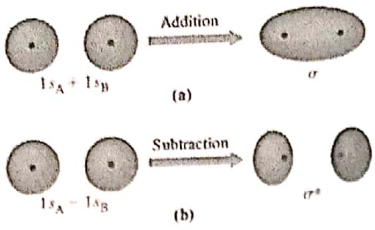


Figure: Formation of bonding and antibonding orbitals (a) The addition of two orbitals in phase to form a molecular orbital. This orbital produces electron density between the nuclei, leading to a chemical bond. (b) The addition of two orbitals out of phase to produce a antibonding orbital. This orbital has a nodal plane perpendicular to the internuclear axis, as do all antibonding orbitals.

Molecular orbital theory

Rules for filling atomic (not molecular) orbitals are:

Aufbau Principle
Place electrons in the ***lowest energy orbitals*** available.

Pauli Exclusion Principle
No orbital may hold more than ***two electrons*** and they must have ***opposite spin***.

Hund's Rule
Every orbital in a subshell must contain only one electron before any orbital can have two electrons. All electrons in singly occupied orbitals must have the same spin.

Molecular orbital theory

Same rules to fill molecular orbitals (not atomic orbital). Let's apply this to the molecular orbital diagram of oxygen.

Each O atom contains 8 electrons, so we have 16 electrons to put into the molecular orbitals of O_2.

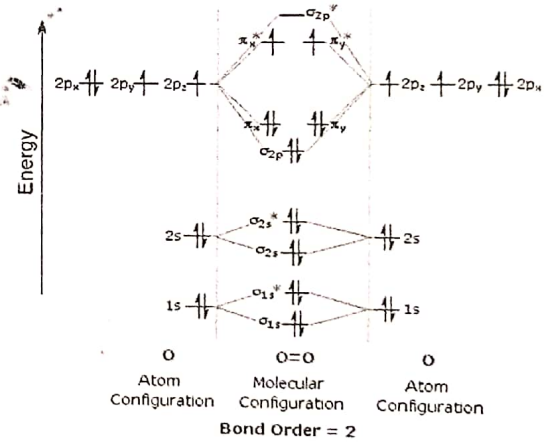
We put two electrons into each of the σ1s, σ1s, *σ2s*, *σ2s*, and σ2p (10 electrons).

The next four electrons follow **Hund's Rule** and go in order into the π_x, π_y, π_x, and π_y orbitals.

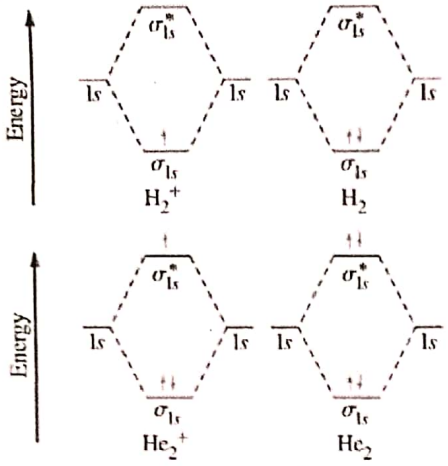
The remaining two electrons follow **Hund's Rule** and go in order into the π^*_x and π^*_y orbitals.

An oxygen molecule has two unpaired electrons.

Molecular orbital theory



2023/02/20 11:02



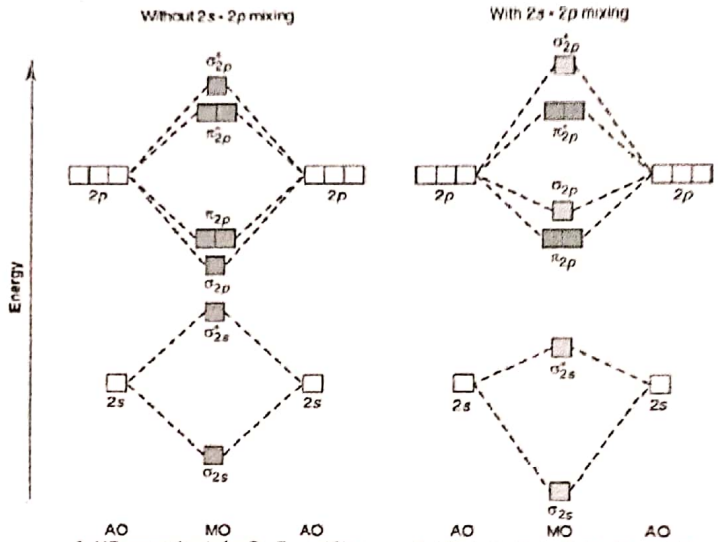
H_2^+ : BO = (1-0)/2 = 1/2

H_2 : BO = (2-0)/2 = 1

He_2^+ : BO = (2-1)/2 = 1/2

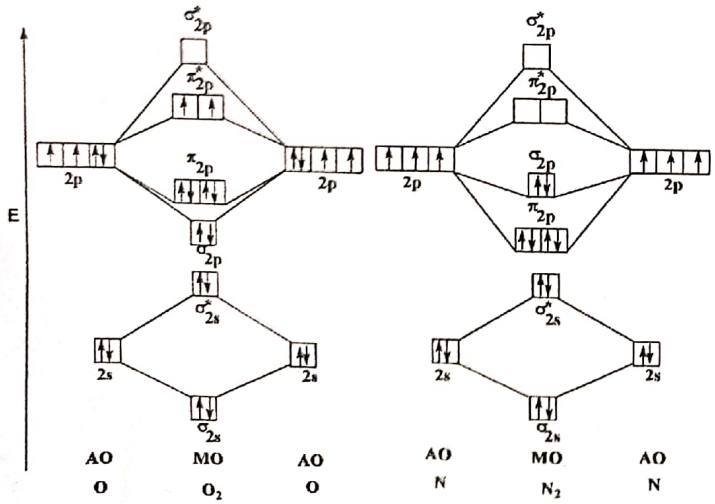
He_2 : BO = (2-2)/2 = 0

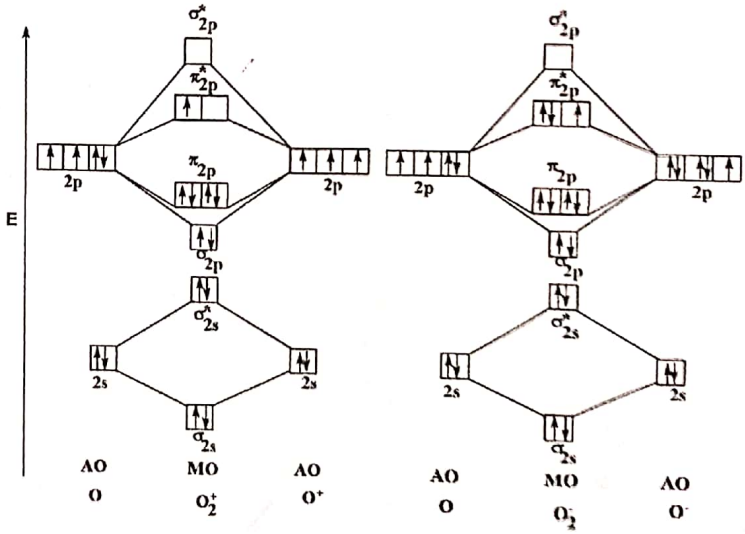
No bond is produced He_2 is not a stable species.



A MO energy levels for O_2, F_2, and Ne_2

B MO energy levels for B_2, C_2, and N_2





2023/02/20 11:02

N_2: σ_{1s}^2 σ_{1s}^{*2} σ_{2s}^2 σ_{2s}^{*2} $\pi_{2p_y}^2$ $\pi_{2p_z}^2$ σ_{2p}^2

$$N_2 \text{ bond order} = \frac{10-4}{2} = 3$$

O_2: σ_{1s}^2 σ_{1s}^{*2} σ_{2s}^2 σ_{2s}^{*2} σ_{2p}^2 $\pi_{2p_y}^2$ $\pi_{2p_z}^2$ $\pi_{2p_y}^{*1}$ $\pi_{2p_z}^{*1}$

$$O_2 \text{ bond order} = \frac{10-6}{2} = 2$$

Predict the stabilities and bond orders of the ions (a) O_2^+ and (b) O_2^-.

(a) We remove one of the π_{2p}^* electrons of O_2 to find the configuration of O_2^+:

σ_{1s}^2 σ_{1s}^{*2} σ_{2s}^2 σ_{2s}^{*2} σ_{2p}^2 $\pi_{2p_y}^2$ $\pi_{2p_z}^2$ $\pi_{2p_y}^{*1}$

Bond order = (10—5)/2 = 2.5

(b) We add one electron to the appropriate orbital of O_2 to find the configuration of O_2^-. Following Hund's Rule, we add this electron into the $\pi_{2p_y}^*$ orbital to form a pair:

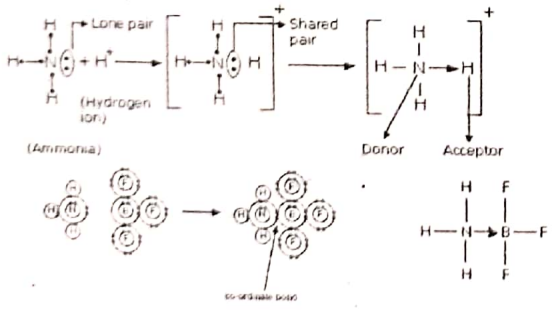
σ_{1s}^2 σ_{1s}^{*2} σ_{2s}^2 σ_{2s}^{*2} σ_{2p}^2 $\pi_{2p_y}^2$ $\pi_{2p_z}^2$ $\pi_{2p_y}^{*2}$ $\pi_{2p_z}^{*1}$

Bond order = (10—7)/2 = 1.5

MOT	VBT
(i) MOs are formed by LCAO approximation method. The AOs involved in the formation of these MOs are from valence shell of two atoms.	(i) Two AOs give an interatomic orbital obtained by the combination of the space functions of two unpaired electrons; one being in each of the two AOs.
(ii) AOs of the resulting molecule lose their individual identity.	(ii) The resulting molecule consists of atoms which, although, interacting, retain their individual character.
(iii) MOT starts with the nuclei of the constituent atoms.	(iii) VBT starts with the individual atoms and consider interaction between them.
(iv) MOT offers a satisfactory explanation of the paramagnetic character of O_2 molecule.	(iv) VBT fails to account for the paramagnetic character of O_2 molecule.
(v) Resonance plays no role in MOT.	(v) Resonance plays an important role in VBT.

Co-ordinate bond

In the formation of a simple covalent bond, each atom supplies one electron to the bond - but that doesn't have to be the case. *A co-ordinate bond (also called a dative covalent bond) is a covalent bond (a shared pair of electrons) in which both electrons come from the same atom.*



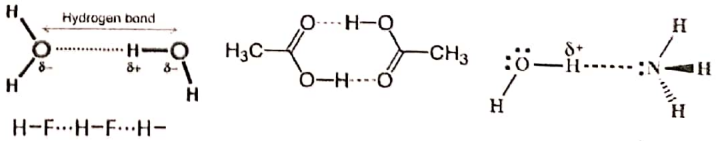
Hydrogen bond

A hydrogen bond is the electrostatic attraction between two polar groups that occurs when a hydrogen (H) atom covalently bound to a highly electronegative atom such as nitrogen (N), oxygen (O), or fluorine (F) experiences the electrostatic field of another highly electronegative atom nearby.

Hydrogen bonding is represented in structural drawings by a series of dots.

—X—H···Y—

Usually, hydrogen bonding is seen in cases where X and Y are the atoms F, O, or N.

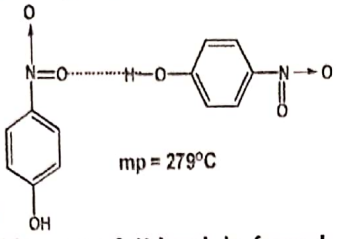


H–F···H–F···H–F

2023/02/20 11:03

Intermolecular and Intra-molecular Hydrogen Bonding

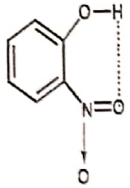
Intermolecular Hydrogen Bonding (Association): This type of H-bond occurs between two or more molecules of the same or different compound. Examples: HF, H_2O, NH_3 para-nitrophenol etc.



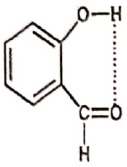
mp = 279°C

H_2O is liquid while H_2S is gas-explain

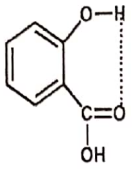
Intra-molecular Hydrogen Bonding: This type of H-bond is formed between a hydrogen atom and an electronegative atom present in the same molecule.



o-nitrophenol (mp = 214°C)



o-hydroxybenzaldehyde



o-hydroxybenzoic acid

2023/02/20 11:03